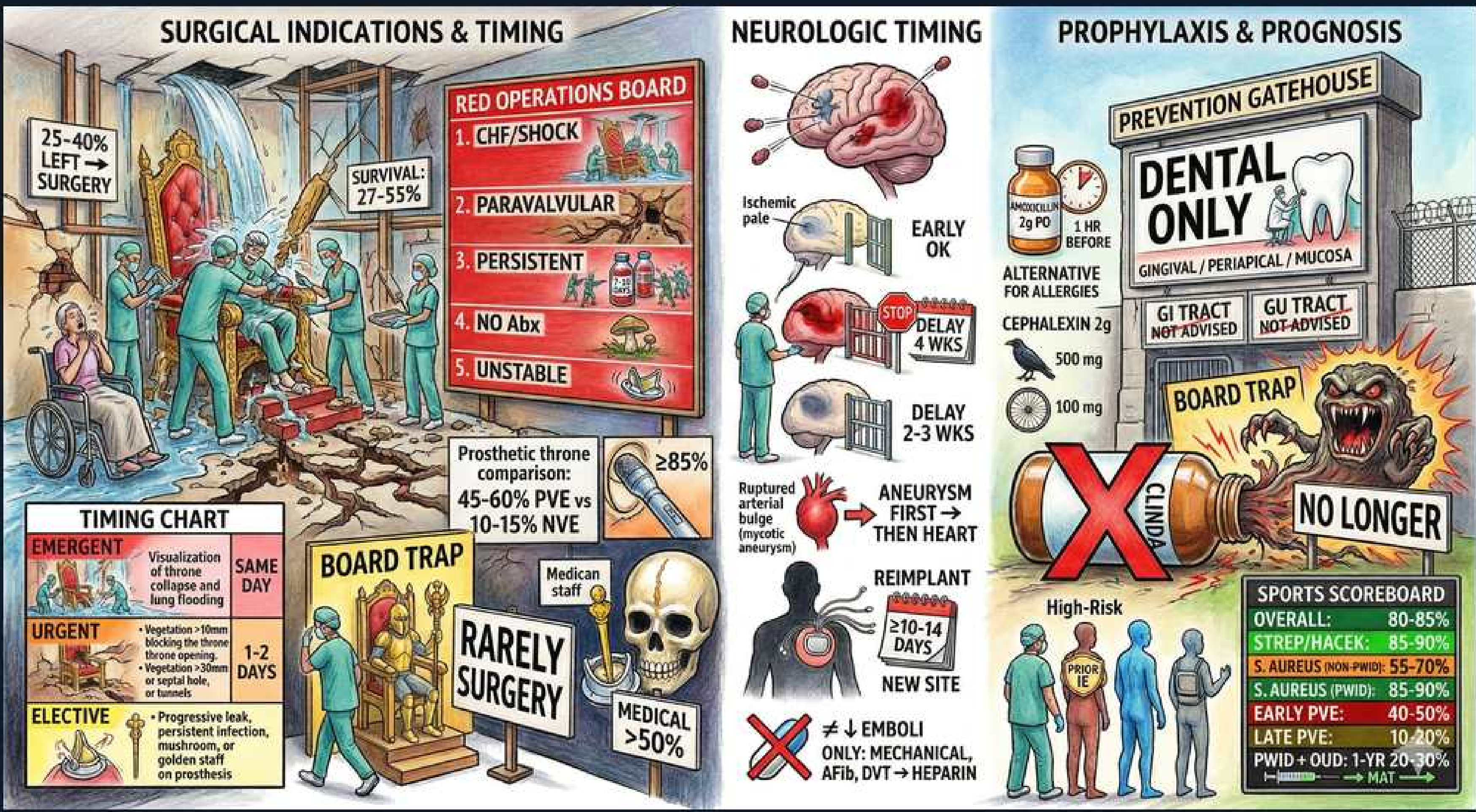
Infective Endocarditis — Surgery, Neuro Timing & Prophylaxis



1. Left-sided surgery rate

WHAT YOU SEE

25-40% left-surgery placard

WHAT IT ENCODES

25-40% of left-sided native-valve IE patients ultimately need surgery during the index hospitalization. Surgery is more common in left-sided than right-sided disease. Drives the board emphasis on early surgical evaluation.

BOARD TRAP

Right-sided IE (IVDU, tricuspid) is usually managed medically — surgery there is the exception, not the rule.

2. CHF / shock — #1 indication

WHAT YOU SEE

Red operations board: CHF/SHOCK

WHAT IT ENCODES

Heart failure from acute valvular regurgitation is the strongest and most common indication for surgery in IE. Acute severe AR or MR causing pulmonary edema mandates urgent/emergent valve repair regardless of antibiotic duration.

BOARD TRAP

Waiting to 'complete antibiotics' before operating a patient in HF is wrong — hemodynamic instability overrides infection-cure timing.

3. Paravalvular / abscess

WHAT YOU SEE

PARAVALVULAR red row

WHAT IT ENCODES

Perivalvular extension (abscess, fistula, new heart block) is a surgical indication. Suspect when PR interval prolongs on serial ECG. TEE is far more sensitive than TTE for detecting abscess.

BOARD TRAP

New AV block in IE = ring abscess until proven otherwise — get a TEE, do not just monitor.

4. Persistent infection

WHAT YOU SEE

PERSISTENT row

WHAT IT ENCODES

Persistent bacteremia or fever >5-7 days despite appropriate antibiotics is a surgical indication, implying uncontrolled focus. Persistent positive cultures suggest abscess or hardware involvement.

BOARD TRAP

Don't keep escalating antibiotics indefinitely — persistent sepsis means source control (surgery) is needed.

5. Difficult organism (No Abx)

WHAT YOU SEE

NO ABX row

WHAT IT ENCODES

Organisms hard to eradicate medically — fungi, multidrug-resistant organisms, Brucella, Q fever — are surgical indications because antibiotics alone rarely cure. Fungal IE almost always needs valve replacement.

BOARD TRAP

Fungal or resistant-organism IE managed medically alone has unacceptably high failure — surgery is part of cure.

6. Unstable / large vegetation

WHAT YOU SEE

UNSTABLE red row

WHAT IT ENCODES

Large mobile vegetations (>10 mm) especially on the mitral valve, and recurrent emboli despite antibiotics, favor early surgery to prevent embolization. Embolic risk is highest in the first 2 weeks.

BOARD TRAP

A vegetation >10 mm WITH an embolic event is a stronger surgical case than size alone.

7. PVE vs NVE survival

WHAT YOU SEE

SURVIVAL 27-55% / PVE comparison

WHAT IT ENCODES

Prosthetic-valve endocarditis carries far worse mortality (45-60% PVE vs 10-15% NVE on the throne comparison). Early PVE (<2 months, often staph/CoNS) is especially lethal and usually surgical.

BOARD TRAP

PVE is not just 'NVE on a prosthesis' — much higher mortality and a much lower surgical threshold.

8. Timing chart — emergent/urgent/elective

WHAT YOU SEE

EMERGENT / URGENT / ELECTIVE timing chart

WHAT IT ENCODES

Emergent (same day) = refractory pulmonary edema/cardiogenic shock. Urgent (1-2 days) = persistent infection, large vegetation, valve dysfunction. Elective (earlier preferred) = paravalvular leak, dehiscence, controlled state.

BOARD TRAP

Stroke does NOT mandate delaying surgery in unstable patients — emergent hemodynamic indications still go same-day.

9. Ischemic stroke — operate

WHAT YOU SEE
EARLY OK / ischemic pale brain

WHAT IT ENCODES
After a non-hemorrhagic (ischemic) embolic stroke without major hemorrhage, valve surgery can proceed early if otherwise indicated — older 'wait 4 weeks' dogma is outdated for small infarcts.

BOARD TRAP
A small ischemic stroke is NOT a reason to delay needed surgery; only large infarcts or bleeding force delay.

10. Hemorrhage — delay 2-4 wks

WHAT YOU SEE
DELAY 4 WKS red brain

WHAT IT ENCODES
Intracranial hemorrhage (or hemorrhagic transformation of a large infarct) requires delaying cardiac surgery ~2-4 weeks because cardiopulmonary bypass heparinization risks catastrophic bleed expansion.

BOARD TRAP
This delay is for HEMORRHAGE, not for routine small ischemic stroke — don't conflate the two.

11. Mycotic aneurysm sequence

WHAT YOU SEE
ANEURYSM FIRST then heart

WHAT IT ENCODES
A ruptured/symptomatic mycotic (infectious) aneurysm should be addressed before cardiac surgery because bypass heparin would worsen bleeding. Unruptured small aneurysms may resolve with antibiotics.

BOARD TRAP
Treat the bleeding mycotic aneurysm FIRST; valve surgery on bypass before securing it can be fatal.

12. Reimplant timing 10-14 days

WHAT YOU SEE
REIMPLANT / 10-14 DAYS / NEW SITE

WHAT IT ENCODES
If a prosthetic valve or device is reimplanted after explant for infection, a clean new site and ~10-14 day window of antibiotic clearance reduces seeding of the new hardware.

BOARD TRAP
Reimplanting into the original infected bed without clearing infection risks immediate re-infection of the new valve.

13. Anticoagulation rule

WHAT YOU SEE
EMBOLI ONLY: mechanical / AFib / DVT heparin

WHAT IT ENCODES
In NVE, anticoagulation is NOT added to reduce embolic risk — it raises intracranial hemorrhage. Continue anticoagulation only for an independent indication (mechanical valve, AFib, DVT/PE), and hold it with CNS hemorrhage.

BOARD TRAP
Starting heparin/warfarin to 'prevent vegetation emboli' is wrong and dangerous — it increases brain bleeding.

14. Prophylaxis — dental only

WHAT YOU SEE
DENTAL ONLY prevention gatehouse

WHAT IT ENCODES
Prophylaxis is limited to high-risk cardiac patients undergoing procedures involving gingival/periapical tissue or oral mucosa perforation. GI and GU procedures are NOT indications.

BOARD TRAP
GI/GU procedures do NOT warrant IE prophylaxis — only high-risk hearts having dental/oral-mucosa manipulation.

15. Amoxicillin 2 g / cephalexin

WHAT YOU SEE
Amoxicillin 2 g PO 1 hr before; cephalexin 2 g

WHAT IT ENCODES
First-line is amoxicillin 2 g PO one hour before the procedure. Penicillin-allergic: cephalexin 2 g (or azithro/clindamycin alternatives), though clindamycin is now de-emphasized.

BOARD TRAP
Clindamycin is NO LONGER recommended as the go-to allergy alternative due to C. difficile risk — cephalexin/azithro preferred.

16. Clindamycin — no longer

WHAT YOU SEE
CLINDA X / NO LONGER

WHAT IT ENCODES
Clindamycin has been removed from routine IE prophylaxis recommendations because of disproportionate C. difficile and adverse-event risk relative to benefit.

BOARD TRAP
Picking clindamycin for the penicillin-allergic patient is now the wrong answer — use a cephalosporin or macrolide.

17. High-risk cardiac groups

WHAT YOU SEE

HIGH-RISK figures / sports scoreboard

WHAT IT ENCODES

Only prosthetic valves/material, prior IE, certain congenital heart disease (unrepaired cyanotic, repaired with residual defect/within 6 months), and cardiac transplant valvulopathy qualify for prophylaxis.

BOARD TRAP

Mitral valve prolapse, bicuspid aortic valve, and HCM are NOT prophylaxis indications — common distractors.